# SRCNet: Seminal Image Representation Collaborative Network for Oil Spill Segmentation in SAR Imagery

Fang Chen, Heiko Balzter, Peng Ren, *Senior Member, IEEE*, and Huiyu Zhou

***Abstract*— Effective oil spill segmentation in synthetic aperture radar (SAR) images is critical for marine oil pollution cleanup, and proper image representation contributes to effective learning for accurate oil spill segmentation. In this article, we propose an effective oil spill segmentation network named SRCNet, which is constructed by leveraging seminal SAR image representation to empower the learning capability of the proposed segmentation network for accurate oil spill segmentation. Specifically, the image representation utilized in our proposed SRCNet originates from SAR imagery, modeling with the internal characteristics of oil spill SAR image data, which therefore promotes effective learning for accurate oil spill segmentation in the training process. Besides, to conduct enhanced oil spill segmentation, we construct the proposed SRCNet with a pair of deep neural nets that work in a competition manner, where one neural net strives to produce accurate oil spill segmentation maps by drawing samples from the collaborated seminal image representation, while the other tries its best to distinguish between the produced and the true segmentations. It is the competition and the image representation collaborated that drives the proposed SRCNet to operate accurate oil spill segmentation efficiently with small amount of training data. This establishes an economical and efficient way for oil spill segmentation. Additionally, to further improve the segmentation performance of the proposed SRCNet, a regularization term that penalizes the segmentation loss is devised, which encourages the produced segmentation to approach the ground-truth segmentation, promoting the segmentation capability of the proposed SRCNet for accurate oil spill segmentation. Experimental evaluations from different metrics validate the effectiveness of the proposed SRCNet for oil spill segmentation.**

***Index Terms*— Collaborative effective learning, marine oil spill segmentation, seminal image representation, synthetic aperture radar (SAR) imagery.**

## I. Introduction

MARINE oil spills are the release of a liquid petroleum hydrocarbon into the ocean waters. The spilled oil at sea normally results in heavy pollution to the marine environment, especially the marine ecosystem [1], [2]. Specifically, oil pollution at sea may have a bad effect on shellfish and other organisms, making them more vulnerable to environmental stressors [3], [4]. Besides, marine oil spills generally occur at different scales, which can cause various damages to the marine environment [5]. Moreover, oil spills on the ocean surface can spread to different square nautical miles in a thin oil slick, and the marine oil spills can also have a great impact on the economy, ecology, and human activities [6], [7]. Therefore, in the operation of marine environmental protection, conducting timely damage assessment and spread control of oil spills is necessary in practice [8], [9], [10]. Satellite-based synthetic aperture radar (SAR) is regarded as a powerful spaceborne sensor for detecting marine oil spills [11], because it has the capabilities of remotely observing environmental and natural targets on the earth's surface in day and night time without depending on weather conditions with wide coverage [12], [13]. Moreover, in the implementation of SAR image segmentation, the construction of the segmentation formulation with image statistical distribution information included contributes to effective segmentation [14]. Therefore, the image statistical distribution is a beneficial component for accurate oil spill segmentation.

Specifically, in the remote sensing community of implementation SAR image segmentation, researchers mainly devoted to investigating physical characteristics that describe SAR image data so as to develop strategies for analyzing oil spills in SAR images. The physical characteristics reflecting surface scattering correlate to the interpretation of ocean backscatter, where the surface scattering types that characterize oil spills in SAR images have been investigated thoroughly. The representative researches from this perspective include those by Salberg et al. [15], Buono et al. [16], Velotto et al. [17], Espeseth et al. [18], and Minchew et al. [19], Collins et al. [20]. They conducted the studies based on the polarimetric properties of SAR image data [21], where the polarimetric properties are explored for facilitating effective oil spill detection. This enables the basic image processing techniques such as graph theory [22] and thresholding [23] methods can be implemented to detect oil spills in SAR images with enhanced representations.

On the other hand, intelligent segmentation methods have been extensively explored for SAR image segmentation [24], [25], [26], where the segmentation with improved performance was addressed [27]. Specifically, to conduct oil spill

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SAR image segmentation, researches from machine learning community construct segmentation strategies mainly based on energy minimization, where an energy functional that measures the similarity and fitness of oil spills is formulated. In this regard, Marques et al. [28] constructed SAR image segmentation energy functional by exploiting the model representation of images in different regions. Xu et al. [29] operated oil spill SAR image segmentation with the segmentation energy functional formulated using an arbitrary pixel within one marine oil spill region as initialization. Xia et al. [30] performed oil spill SAR image segmentation with the integration of the non-local interactions between patch pairs to formulate the segmentation functional. Chen et al. [31] utilized the alternating direction method of multipliers to construct the segmentation energy functional, by which the constructed segmentation method is capable of performing oil spill segmentation from blurry SAR images. Jing et al. [32] utilized global minimization active contour model to define the segmentation energy functional, and this enables oil spill detection without local minima. Moreover, to address the segmentation of oil spill SAR images in terms of precisely preserving oil spill edges in the segmentation results, Chen et al. [33] presented an edge sensitive segmentation scheme in which the self-guided filtering is incorporated in the construction of the segmentation energy functional. These strategies operate oil spill image segmentation by fitting region information with active contour models, where the segmentation operation requires manually labeling an initial region for starting the segmentation. In this scenario, the segmentation output is correlated to the labeled initialization region to some extent, and a proper initialization region labeled contributes to a more accurate segmentation [34]. Particularly, in SAR imagery for observation maritime scenes, strong wind can generate rough sea surfaces, but spread oil slicks quickly and create clutter [35]. Besides, turbulence and wind blowing on the ocean surface can result in oil spills in SAR images exhibit various areas with irregular shapes. This makes difficulty for manually labeling in terms of capturing a reliable initialization region for stable and accurate oil spill segmentation.

Additionally, in image segmentation, the deep learning-based segmentation techniques have been developed for implementation oil spill segmentation in SAR images [36], [37], [38], [39]. In this scenario, Li et al. [40] operated oil spill detection with multiscale conditional adversarial networks. Ma et al. [41] performed oil spill detection by using polarimetric scattering information to improve the detection accuracy. Kang et al. [42] performed oil spill segmentation with self-supervised spectral–spatial transformer network. Li et al. [43] constructed a self-evolving deep learning algorithm for oil spill detection in SAR images. Besides, De Laurentiis et al. [44] investigated slick classification within a deep learning framework to assess the capabilities of deep architectures for providing an accurate classifier for separating oil slicks from SAR images. Furthermore, to address the inherent speckle noise in SAR images, Qian et al. [45] proposed a deep neural network, which combines the speckle reduction regularization for SAR image scene classification. The deep learning-based segmentation methods perform oil spill segmentation automatically. However, the successful training of a deep neural segmentation model generally requires large amounts of training data with high computations, which is neither economical nor efficient in practical segmentation implementation. Therefore, in the operation of oil spill SAR image segmentation, to render accurate segmentation, we develop an efficient and effective segmentation network by leveraging seminal SAR image representation to empower effective learning for accurate oil spill segmentation.

Particularly, to perform effective oil spill SAR image segmentation, we propose a seminal image representation collaborative segmentation network named SRCNet, which is constructed by considering SAR image formation and the training for oil spill segmentation comprehensively. The segmentation work [34] operated oil spill segmentation based on energy minimization, in which the representation of oil spill SAR image data was not explored. In contrast, the proposed SRCNet operates oil spill segmentation by leveraging seminal SAR image representation to facilitate its learning capability for effective oil spill segmentation. The seminal image representation utilized originates from SAR imagery, modeling with the internal characteristics of the oil spill SAR image data, which therefore empowers the learning capability of the proposed SRCNet for performing accurate oil spill segmentation. Besides, to operate oil spill segmentation with enhanced performance, the proposed SRCNet is constructed with a pair of deep neural nets that work in a competition manner, with one deep neural net strives to produce accurate oil spill segmentation maps by drawing samples from the collaborated seminal image representation, while the other tries its best to distinguish between the produced segmentation and the real oil spill segmentation. The competition and the seminal image representation collaborated drive the proposed model to operate accurate oil spill segmentation efficiently without requiring a large amount of training data. Furthermore, to further improve the segmentation performance of the proposed SRCNet, a regularization term that penalizes the segmentation loss is introduced. In the training process, the segmentation loss regularization term encourages the produced segmentation to approach the ground-truth segmentation, promoting the segmentation capability of the proposed segmentation network for a more accurate oil spill segmentation. The main contributions of this work are summarized as follows.

1) We propose a novel marine oil spill SAR image segmentation network named SRCNet, which is structured with a pair of deep neural nets running in a competitive manner, in which the seminal image representation that characterizes oil spill SAR image data is collaborated. The collaboration of competition and the seminal image representation enables to perform effective oil spill segmentation.
2) We operate the training for these two deep neural nets simultaneously, and the collaborated seminal image representation empowers effective learning for accurate oil spill segmentation at a fast speed with a small amount of training image data. This establishes an efficient

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and economical way for operating oil spill SAR image segmentation.

3) We devise a segmentation loss regularization term and introduce it into our proposed segmentation model to penalize the loss between the produced segmentation and the ground-truth segmentation. This improves the segmentation capability of our proposed SRCNet for performing a more accurate oil spill segmentation in SAR images.

Extensive experimental evaluations from different metrics validate the effectiveness of our proposed SRCNet for oil spill segmentation over different sources of SAR images.

## II. Related Work

Deep architectures, which have the capability of learning rich models to represent data encountered in intelligent tasks such as image processing, are extensively explored [46]. Besides, for marine oil spill SAR image segmentation, to conduct stable and accurate segmentation, automatic and effective segmentation methods are pursued. Thus, to implement effective oil spill segmentation in SAR images automatically, we construct an oil spill segmentation network by taking the advantage of recent advances of deep neural networks in terms of training for image segmentation, in which image representation is utilized for effective segmentation. Particularly, in the application of deep neural networks for image segmentation, a striking point is the involvement of discriminative models [47]. Therefore, to present the procedures of constructing the proposed segmentation network, we here briefly review a representative deep neural network that performs image segmentation by pairing the generative model with a discriminative model. Specifically, let $p_{\text{data}}(\mathbf{x})$ be the distribution of the input data, $p_{\mathbf{z}}(\mathbf{z})$ be the prior distribution of the noise variables $\mathbf{z}$, $p_{\text{gen}}(\mathbf{x})$ be the learned distribution of the generative model over data $\mathbf{x}$, $D(\mathbf{x})$ be the probability that the discriminative net obtaining $\mathbf{x}$ from data rather than $p_{\text{gen}}$, and $G(\mathbf{z})$ represents the probability of mapping the input variables $\mathbf{z}$ to data space. These two models thus build a value function shown as follows:

$$\begin{aligned}\mathcal{V}(G,D) &= \int p_{\text{data}}(\mathbf{x})\log\big(D(\mathbf{x})\big)dx \\ &\quad + \int p_{\mathbf{z}}(\mathbf{z})\log\big(1-D(G(\mathbf{z}))\big)dz \\ &= \int p_{\text{data}}(\mathbf{x})\log\big(D(\mathbf{x})\big) \\ &\quad + p_{\text{gen}}(\mathbf{x})\log\big(1-D(\mathbf{x})\big)dx. \end{aligned} \tag{1}$$

This formulation shows the correlation between different distributions for operating image segmentation. In the segmentation implementation, the objective is to learn a well-represented distribution $p_{\text{gen}}(\mathbf{x})$ so as to produce accurate segmentations. For clarity, we first rewrite (1) as follows:

$$\begin{aligned}\mathcal{V}(G,D) &= \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}(\mathbf{x})}\big[\log D(\mathbf{x})\big] \\ &\quad + \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}(\mathbf{z})}\big[\log(1-D(G(\mathbf{z})))\big]. \end{aligned} \tag{2}$$

This equation represents the value function using the expectation form where it clearly demonstrates that $\log D(\mathbf{x})$ and $\log(1-D(G(\mathbf{z})))$ are computed over $p_{\text{data}}(\mathbf{x})$ and $p_{\mathbf{z}}(\mathbf{z})$, respectively. According to (1) and (2), we can observe that the generative model learns to produce segmentation outputs by drawing samples from $p_{\mathbf{z}}(\mathbf{z})$, while the discriminative model learns to determine whether a segmentation output is from the data distribution $p_{\text{data}}$ or the model distribution $p_{\text{gen}}$. Thus, the training for these two models is conducted by minimizing $\log(1-D(G(\mathbf{z})))$ for generating accurate segmentations, and in the meantime maximizing the probability of the discriminator for assigning the correct label to samples from both real image data and the generator. They thus build a two-player mini-max game, which is shown as follows:

$$\begin{aligned}\min_G \max_D \ \mathcal{V}(G,D) &= \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}(\mathbf{x})}\big[\log D(\mathbf{x})\big] + \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}(\mathbf{z})} \\ &\quad \times \big[\log(1-D(G(\mathbf{z})))\big]. \end{aligned} \tag{3}$$

Examining this formulation, it is obvious that the inner loop is to maximize the value function with respect to the discriminator $D$, whereas the outer loop is to minimize the value function with respect to the generator $G$. The convergence achieves when the discriminator is unable to distinguish a segmentation that comes from the generative model or the real image data. Therefore, optimizing the training criterion for improving the capability of $D$ benefits to a more effective segmentation. Equation (3) shows that the training criterion for $D$, given any $G$, is to maximize the quantity of the value function $\mathcal{V}(G,D)$. In combination with (1), the optimal training criterion for $D$ in (3) can be reformulated as follows:

$$\begin{aligned} C(G) &= \max_D \ \mathcal{V}(G,D) \\ &= \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}}\big[\log D^*(\mathbf{x})\big] + \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}}\big[\log(1-D^*(G(\mathbf{z})))\big] \\ &= \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}}\big[\log D^*(\mathbf{x})\big] + \mathbb{E}_{\mathbf{x}\sim p_{gen}}\big[\log(1-D^*(\mathbf{x}))\big]. \end{aligned} \tag{4}$$

In the training process, the generative model is trained to produce segmentations that the discriminator is unable to differentiate, i.e., $p_{\text{gen}}(\mathbf{x})$ equals to $p_{\text{data}}(\mathbf{x})$, in which case the minimization computation for $G$ is achieved. From the roles of the generative model and the discriminative model, and the training criterion for them, it is obvious that the training efficiency and the segmentation performance of the segmentation network depend highly on the performance of the generative model in terms of the speed of producing segmentations that as accurate as the segmentations from real image data. Therefore, we aim to construct an effective oil spill segmentation network in which the generative net is encouraged to produce accurate oil spill segmentation maps in high efficiency.

## III. Construction of Seminal Representation Collaborative Oil Spill Segmentation Network

To conduct accurate oil spill segmentation in SAR images in an automatic way, we propose an effective oil spill segmentation network by taking the recent advances of deep neural networks in terms of training for image segmentation without requiring a large amount of training image data. Particularly, in the application of deep neural networks for

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image processing, one promise is to learn models that represent the distribution representations over data encountered in the implementation tasks [46]. Thus, to render accurate oil spill segmentation, in this section, we construct the proposed oil spill segmentation network by leveraging seminal image representation that characterizes oil spill SAR image data to empower its learning capability for operation efficient and accurate oil spill segmentation. The details of the proposed segmentation network are described in the following.

### A. Seminal Representation of Marine Oil Spill SAR Images

We operate oil spill segmentation in SAR images by constructing a seminal image representation collaborative segmentation network named SRCNet, which is capable of performing effective learning for accurate oil spill segmentation efficiently. To present the details of the construction of the proposed segmentation network, we here investigate the seminal image representation of oil spill SAR image data. According to the exploration of SAR imagery for observation of oceanic scenes and the established distribution representation for SAR image data [48], [49], we obtain the distribution representation that characterizes oil spill SAR image data shown as follows:

$$\mathcal{P}_{\text{sedi}}(\mathcal{I}(\mathbf{x})) = \frac{1}{K_s \sigma(\mathbf{x})} \exp^{\left(-\frac{1}{K_s \sigma(\mathbf{x})} \mathcal{I}(\mathbf{x})\right)}, \quad \mathcal{I}(\mathbf{x}) \geq 0 \tag{5}$$

where $I(\mathbf{x})$ represents SAR image vector, $K_s$ is the detection system constant, and $\sigma$ is the normalized radar cross section component. Particularly, in SAR imagery for oil spill depiction, the appearance of oil spill areas in SAR images is correlated to $\sigma$, the power of backscattered radar signals [12], [50]. Besides, in the distribution representation, the component $\sigma$ is information bearing of SAR image pixels [49], and thus it carries the information of each image pixel. Therefore, (5) models the seminal image distribution representation that characterizes oil spill SAR image data, and it thereby provides internal feature information for detecting oil spills in SAR images. Thus, in the implementation of oil spill SAR image segmentation, constructing of the segmentation network with the seminal image representation utilized contributes to effective learning for accurate segmentation.

### B. Construction of Seminal Image Representation Collaborative Network for Oil Spill Segmentation

We construct our proposed SRCNet by taking the success of deep learning, where two deep neural nets running in different roles are structured for enhanced oil spill segmentation in an end-to-end manner, and the training for these two nets is conducted simultaneously. Particularly, in the implementation of image segmentation using deep neural networks, a proper image representation utilized benefits to improve segmentation performance [51]. Thus, to render effective oil spill SAR image segmentation, in our proposed SRCNet, the seminal image representation that characterizes oil spill SAR image data is incorporated. The seminal image representation models with the internal characteristics of oil spill SAR image data, which therefore empowers effective learning for accurate oil spill segmentation in the training process. The seminal image representation of oil spill SAR image data is illustrated in Section III-A. We next provide the detailed processes of constructing the proposed oil spill segmentation network.

Specifically, to achieve enhanced oil spill segmentation, in the proposed segmentation network, a pair of deep neural nets that work in a competition manner are structured, where one deep neural net is structured as mapped generative net, which strives to produce accurate oil spill segmentation maps by drawing samples from the collaborated seminal image representation, while the other is the discriminative net, which tries its best to distinguish between the generated segmentation and the ground-truth segmentation. In this regard, in the training for oil spill image segmentation, when the generated oil spill segmentation maps are as accurate as the ground-truth segmentations, in which case the discriminative net is not able to differentiate these two segmentations, enabling the segmentation network reaches convergence. Thus, the segmentation performance of the proposed segmentation network is closely correlated to the mapped generative net in terms of producing accurate oil spill segmentation maps. In this regard, in our proposed SRCNet, benefiting from the seminal image representation collaborated, the structured mapped generative net produces accurate oil spill segmentation maps at a fast speed, empowering the capability of the proposed SRCNet in terms of performing efficient and accurate oil spill segmentation. Besides, to circumvent the information rough alignment from the input to the generated output [52], we construct the mapped generative net with the mirrored deep neural layers connected to enable the desired low level image information shuttled directly across the net for operation a more detailed oil spill segmentation. Specifically, to represent the interaction between the mapped generative net and the discriminative net for operation oil spill segmentation, let $\mathcal{P}_{\text{sedi}}(\mathbf{x})$ be the seminal image representation of oil spill SAR image data, $\mathcal{G}_{\text{gedi}}(\mathbf{y})$ be the distribution representation of the generated segmentation outputs, and $\mathcal{G}_{\text{gtdi}}(\mathbf{y})$ be the distribution representation of the ground-truth segmentations. Supposing the segmentation is operated over image domain $\Omega$. In the training for oil spill image segmentation, the loss measurement is given as follows:

$$\begin{aligned}\mathcal{L}_{\text{seg}}(G, D) &= \int \mathcal{G}_{\text{gtdi}}(\mathbf{y}) \log\left(\mathcal{D}(\mathbf{y})\right) dy \\ &\quad + \int \mathcal{P}_{\text{sedi}}(\mathbf{x}) \log\left(1 - D(G(\mathbf{x}))\right) dx \\ &= \mathbb{E}_{\mathbf{y}\sim\mathcal{G}_{\text{gtdi}}}\left[\log D(\mathbf{y})\right] \\ &\quad + \mathbb{E}_{\mathbf{x}\sim\mathcal{P}_{\text{sedi}}}[\log\left(1 - D(G(\mathbf{x}))\right)].\end{aligned} \tag{6}$$

This formulation builds the correlation between the mapped generative and the discriminative components. Particularly, in oil spill SAR image representation, the oil spills in SAR images normally exhibit various areas with irregular shapes [35]. This results in difficulty in accurate oil spill segmentation. To address this, in the operation of oil spill segmentation, to improve the segmentation performance of our proposed SRCNet in terms of accurately delineating oil spill areas in SAR images, a regularization term that penalizes the segmentation loss is formulated. In the training process, the

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segmentation loss regularization term encourages the produced segmentation to approach the ground-truth segmentation, promoting the segmentation capability of our proposed SRCNet for accurate oil spill segmentation. The regularization term is given below

$$\mathcal{L}_{\text{sreg}}(G) = \mathbb{E}_{\mathbf{x}\sim\mathcal{P}_{\text{sedi}},\mathbf{y}\sim\mathcal{G}_{\text{gedi}}}[\parallel \mathbf{y} - G(\mathbf{x}) \parallel_2]. \tag{7}$$

This formulation measures the loss of training for generating oil spill segmentation maps. Therefore, incorporating this loss regularization term into the proposed segmentation model contributes to improving the segmentation performance of the proposed segmentation network. Combining (6) and (7), we formulate the joint loss function shown as follows:

$$\begin{aligned}\mathcal{L}_{\text{joint}} &= \gamma_{\text{seg}}\mathcal{L}_{\text{seg}}(G, D) + \gamma_{\text{sreg}}\mathcal{L}_{\text{sreg}}(G) \\ &= \gamma_{\text{seg}}\left[\mathbb{E}_{\mathbf{y}\sim\mathcal{G}_{\text{gtdi}}}\left[\log D(\mathbf{y})\right] + \mathbb{E}_{\mathbf{x}\sim\mathcal{P}_{\text{sedi}}}[\log\left(1 - D(G(\mathbf{x}))\right)]\right] \\ &\quad + \gamma_{\text{sreg}}\mathbb{E}_{\mathbf{x}\sim\mathcal{P}_{\text{sedi}},\mathbf{y}\sim\mathcal{G}_{\text{gedi}}}[\parallel \mathbf{y} - G(\mathbf{x}) \parallel_2]\end{aligned} \tag{8}$$

where $\gamma_{\text{seg}}$ and $\gamma_{\text{sreg}}$ are weight coefficients that balance the loss terms for effective oil spill segmentation. Equation (8) formulates the objective function correlating the two nets $G$ and $D$ in our proposed segmentation network. In the implementation of oil spill SAR image segmentation, to achieve optimal oil spill segmentation, we optimize the objective function with respect to $G$ and $D$ shown as follows:

$$\min_G \max_D \ \gamma_{\text{seg}}\mathcal{L}_{\text{seg}}(G, D) + \gamma_{\text{sreg}}\mathcal{L}_{\text{sreg}}(G). \tag{9}$$

The proposed segmentation network is constructed with a pair of deep neural nets that work in a competitive manner, in which the seminal image representation that characterizes oil spill SAR image data is collaborated to empower effective learning for accurate oil spill segmentation. The training for these two nets is conducted simultaneously to achieve automatic oil spill segmentation. The framework of the proposed SRCNet is illustrated in Fig. 1. Specifically, in this proposed segmentation network, the basic components are the encoder and the decoder, and the detailed structures of these two modules are shown in Fig. 2. In this figure, it is clear that the encoder is composed of a convolutional layer followed by a Batchnorm layer and an LReLU layer. The decoder has an architecture that is similar to the encoder, with the convolutional layer replaced with a deconvolutional layer.

## IV. Implementation Training for Oil Spill SAR Image Segmentation

We propose a seminal image representation collaborative segmentation network named SRCNet, which performs marine oil spill segmentation by leveraging seminal SAR image representation to empower efficient learning for accurate segmentation. The details of constructing the proposed segmentation network are presented in Section III. Specifically, in the proposed segmentation network, two deep neural nets are structured that operate in a competitive manner, where one is the mapped generative net, which strives to produce accurate oil spill segmentation maps by drawing samples from the seminal image distribution representation, while the other is the discriminative net, which aims to distinguish segmentations that come from the mapped generative net or the ground-truth segmentation. They thus build a min-max two-player game in the training process, and convergence is achieved when the generated segmentations are as accurate as the ground-truth segmentations, in which case the discriminative net is not able to distinguish. To facilitate the training process, we represent the optimization for the min-max operation as follows:

$$\begin{aligned}&\min_G \max_D \ \mathcal{L}_{\text{joint}}(G, D) \\ &= \min_G \max_D \ \gamma_{\text{seg}}\mathcal{L}_{\text{seg}}(G, D) + \gamma_{\text{sreg}} \\ &\mathcal{L}_{\text{sreg}}(G) \\ &= \min_G \max_D \ \Big\{\gamma_{\text{seg}}\left[\mathbb{E}_{\mathbf{y}\sim\mathcal{G}_{\text{gtdi}}}\left[\log D(\mathbf{y})\right]\right. \\ &\qquad \left. + \mathbb{E}_{\mathbf{x}\sim\mathcal{P}_{\text{sedi}}}[\log\left(1 - D(G(\mathbf{x}))\right)]\right] \\ &\qquad + \gamma_{\text{sreg}}\mathbb{E}_{\mathbf{x}\sim\mathcal{P}_{\text{sedi}},\mathbf{y}\sim\mathcal{G}_{\text{gedi}}}[\parallel \mathbf{y} - G(\mathbf{x}) \parallel_2]\Big\}.\end{aligned} \tag{10}$$

Particularly, to achieve effective oil spill segmentation, we perform pixel-wise segmentation for oil spill images, enabling that the pixels in the image domain are correctly segmented. Thus, for an input oil spill SAR image $\mathcal{I}(\mathbf{x})$ with $N$ data points in the image domain $\Omega$ (i.e., $\mathcal{I}(\mathbf{x}) = \{\mathcal{I}(\mathbf{x}^{(i)})\}_{i=1}^{N}$), we represent the objective function in (10) with respect to individual data point as follows:

$$\begin{aligned}&\min_G \max_D \ \mathcal{L}_{\text{joint}}(G, D) \\ &= \min_G \max_D \ \Big\{\gamma_{\text{seg}}\left[\mathbb{E}_{\mathbf{y}^{(i)}\sim\mathcal{G}_{\text{gtdi}}}\left[\log D(\mathbf{y}^{(i)})\right]\right. \\ &\qquad \left. + \mathbb{E}_{\mathbf{x}^{(i)}\sim\mathcal{P}_{\text{sedi}}}[\log\left(1 - D(G(\mathbf{x}^{(i)}))\right)]\right] + \gamma_{\text{sreg}} \\ &\qquad \times \mathbb{E}_{\mathbf{x}^{(i)}\sim\mathcal{P}_{\text{sedi}},\mathbf{y}^{(i)}\sim\mathcal{G}_{\text{gedi}}}[\parallel \mathbf{y}^{(i)} - G(\mathbf{x}^{(i)}) \parallel_2]\Big\}.\end{aligned} \tag{11}$$

The application of deep learning for intelligent tasks is to learn models that represent the distribution representations over the employed data [46], in which different tasks are characterized by different parameters. In our proposed SRCNet, two deep neural nets are structured to play different roles in the segmentation work, where the mapped generative net $G$ is structured to produce accurate oil spill segmentation maps, and the discriminative net $D$ is structured to discriminate the generated segmentation from the ground-truth segmentation. Thus, in the implementation of oil spill SAR image segmentation with the proposed SRCNet, we exploit $\varphi_{\text{ptg}}$ and $\varphi_{\text{ptd}}$ to denote the parameters of the components $G$ and $D$, respectively. In the training process, to learn optimal models for describing the segmentation work, we optimize $\varphi_{\text{ptg}}$ and $\varphi_{\text{ptd}}$ rather than $G$ and $D$, and we thus rewrite (11) as follows:

$$\begin{aligned}&\min_{\varphi_{\text{ptg}}} \max_{\varphi_{\text{ptd}}} \ \mathcal{L}_{\text{joint}}(G, D) \\ &= \min_{\varphi_{\text{ptg}}} \max_{\varphi_{\text{ptd}}} \ \Big\{\gamma_{\text{seg}}\left[\mathbb{E}_{\mathbf{y}^{(i)}\sim\mathcal{G}_{\text{gtdi}}}\left[\log D(\mathbf{y}^{(i)})\right]\right. \\ &\qquad \left. + \mathbb{E}_{\mathbf{x}^{(i)}\sim\mathcal{P}_{\text{sedi}}}[\log\left(1 - D(G(\mathbf{x}^{(i)}))\right)]\right] + \gamma_{\text{sreg}} \\ &\qquad \times \mathbb{E}_{\mathbf{x}^{(i)}\sim\mathcal{P}_{\text{sedi}},\mathbf{y}^{(i)}\sim\mathcal{G}_{\text{gedi}}}[\parallel \mathbf{y}^{(i)} - G(\mathbf{x}^{(i)}) \parallel_2]\Big\}.\end{aligned} \tag{12}$$

This formulation shows that the optimization for $\varphi_{\text{ptd}}$ and $\varphi_{\text{ptg}}$ is conducted with respect to the maximization operation and the minimization operation, respectively. Examining this

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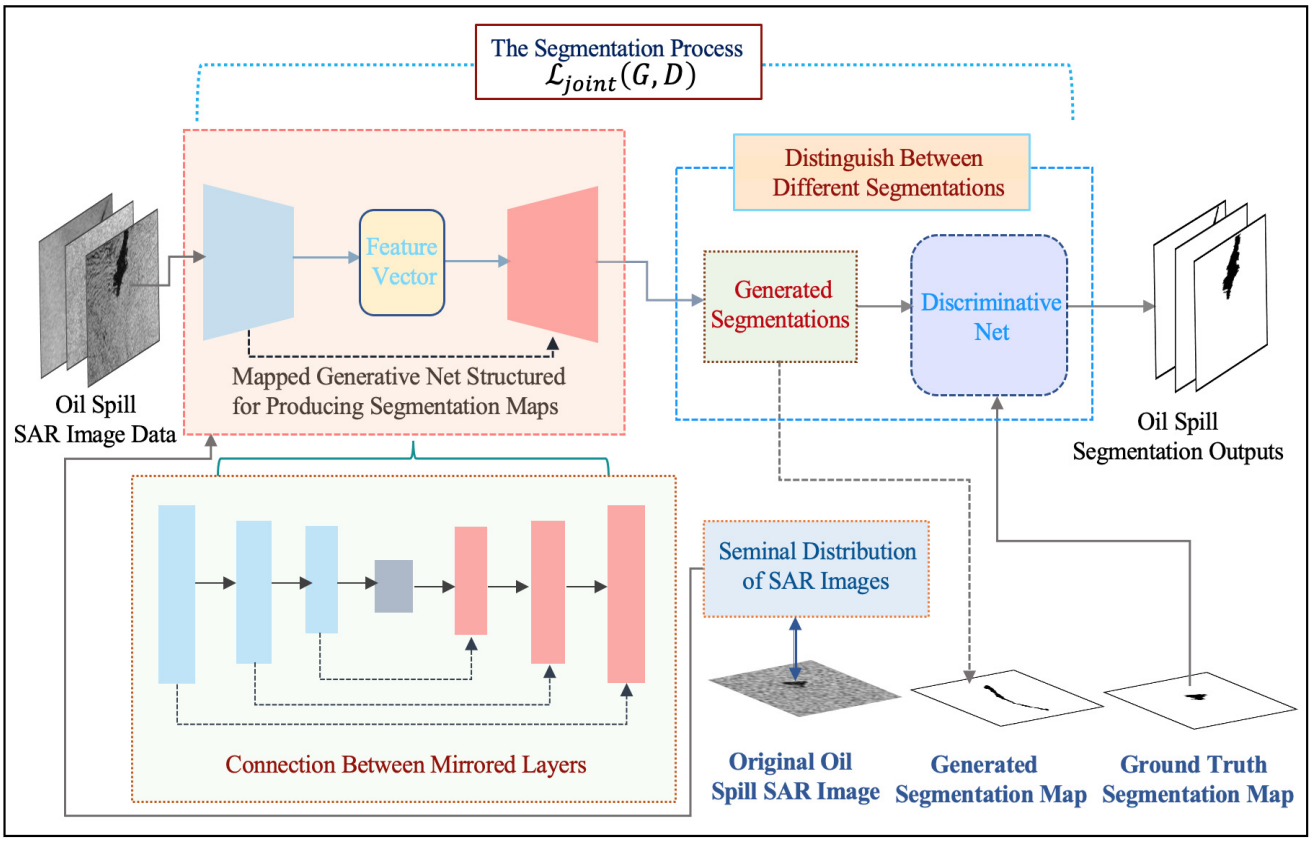


Fig. 1. Architecture of our proposed oil spill SAR image segmentation network. Specifically, the proposed segmentation network is constructed with a mapped generative neural net and a discriminative net working in a competitive manner, with the mapped generative net structured to produce accurate oil spill segmentation maps by drawing samples from the seminal image distribution representation of SAR images, while the discriminative net is trained to distinguish between the generated segmentation and the ground-truth segmentation.

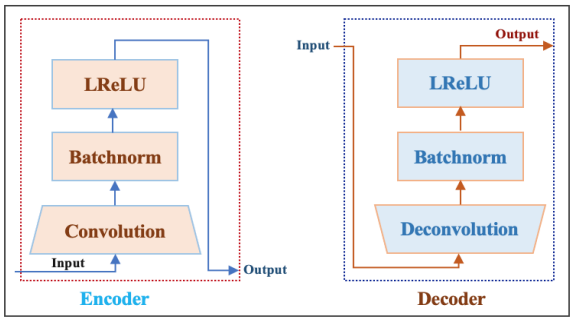


Fig. 2. Structures of the encoder and decoder that consist of the basic components of the oil spill SAR image segmentation network.

equation, we represent the optimization in terms of the maximization step given as follows:

$$\max_{\varphi_{\text{ptd}}} \mathcal{L}_{\text{joint}}(G, D) = \max_{\varphi_{\text{ptd}}} \gamma_{\text{seg}}\left[\mathbb{E}_{\mathbf{y}^{(i)}\sim\mathcal{G}_{\text{gtdi}}}\left[\log D(\mathbf{y}^{(i)})\right] + \mathbb{E}_{\mathbf{x}^{(i)}\sim\mathcal{P}_{\text{sedi}}}[\log\left(1 - D(G(\mathbf{x}^{(i)}))\right)]\right] \quad (13)$$

and the minimization step given as follows:

$$\min_{\varphi_{\text{ptg}}} \mathcal{L}_{\text{joint}}(G, D) = \min_{\varphi_{\text{ptg}}} \gamma_{\text{seg}}\mathbb{E}_{\mathbf{x}^{(i)}\sim\mathcal{P}_{\text{sedi}}}[\log\left(1 - D(G(\mathbf{x}^{(i)}))\right)] + \gamma_{\text{sreg}}\mathbb{E}_{\mathbf{x}^{(i)}\sim\mathcal{P}_{\text{sedi}},\mathbf{y}^{(i)}\sim\mathcal{G}_{\text{gedi}}}[\| \mathbf{y}^{(i)} - G(\mathbf{x}^{(i)}) \|_2]. \quad (14)$$

In the proposed segmentation network, the mapped generative net generates oil spill segmentation maps by drawing samples from the seminal image distribution representation $\mathcal{P}_{\text{sedi}}(\mathbf{x})$, while the discriminative net differentiates the generated segmentation from the ground-truth segmentation also based on samples. Therefore, in the training process, given a finite training set, the expectations in (13) and (14) can be approximated using a minibatch of samples, and the loss functions of these two nets are given as follows:

$$\mathcal{L}_{L_{\text{seg}}} = \frac{1}{m}\sum_{i=1}^{m}\gamma_{\text{seg}}\left[\log D(\mathbf{y}^{(i)}) + \log\left(1 - D(G(\mathbf{x}^{(i)}))\right)\right] \quad (15)$$

and

$$\mathcal{L}_{L_{\text{gen}}} = \frac{1}{m}\sum_{i=1}^{m}[\gamma_{\text{seg}}\log\left(1 - D(G(\mathbf{x}^{(i)}))\right) + \gamma_{\text{sreg}}[\| \mathbf{y}^{(i)} - G(\mathbf{x}^{(i)}) \|_2]] \quad (16)$$

where $m$ is the size of minibatch. Thus, the optimization for the segmentation network is achieved by alternatively optimizing the parameters of the deep neural nets $G$ and $D$ with respect to their loss functions. Particularly, the proposed SRCNet performs accurate oil spill segmentation with

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**Algorithm 1** Procedures of Training the Proposed Segmentation Method for Marine Oil Spill SAR Image Segmentation

**Input**: Marine oil spill image dataset including both the original images and the ground-truth segmentations.

**Output**: Oil spill segmentation maps.

**for** in the training iteration numbers **do**

Optimisation for the maximisation step

- Draw mini-batch of $m$ examples from the prior $\mathcal{P}_{sedi}(\mathbf{x})$;
- Draw mini-batch of $m$ examples from the ground truth distribution $\mathcal{G}_{gtdi}(\mathbf{y})$;
- Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\varphi_{ptd}} \frac{1}{m} \sum_{i=1}^{m} \gamma_{seg} \left[ \log D(\mathbf{y}^{(i)}) + \log \left(1 - D(G(\mathbf{x}^{(i)}))\right)\right];$$

Optimisation for the minimisation step

- Draw mini-batch of $m$ examples from the prior $\mathcal{P}_{sedi}(\mathbf{x})$;
- Draw mini-batch of $m$ examples from the ground truth distribution $\mathcal{G}_{gtdi}(\mathbf{y})$;
- Update the mapped generative net by descending its stochastic gradient:

$$\nabla_{\varphi_{ptg}} \frac{1}{m} \sum_{i=1}^{m} [\gamma_{seg} \log \left(1 - D(G(\mathbf{x}^{(i)}))\right) + \gamma_{sreg} [\| \mathbf{y}^{(i)} - G(\mathbf{x}^{(i)}) \|_2]];$$

**end for**;

a small amount of image training data. This benefits from the fact that the proposed SRCNet is constructed with the collaboration of the seminal image representation, by which the internal characteristics of the image data are modeled to instruct the mapped generative net producing accurate oil spill segmentations. Additionally, to further increase the segmentation performance of the proposed SRCNet, a regularization term that penalizes the segmentation loss is formulated, which encourages the generated segmentation to be close to the ground-truth segmentation, promoting the segmentation capability of the proposed segmentation network for accurate oil spill segmentation. The proposed segmentation network provides an economical way for oil spill image segmentation, where the availability of oil spill SAR image data is limited in practice. The illustration of the proposed segmentation network for oil spill segmentation is described in Algorithm 1.

## V. Experimental Work

We have presented the details of constructing our proposed SRCNet in Sections III and IV. To demonstrate the effectiveness of our proposed SRCNet for oil spill SAR image segmentation, in this section, we perform the experimental work by comparing its segmentation performance against several state-of-the-art segmentation methodologies. Particularly, to conduct a more comprehensive evaluation for the proposed SRCNet, the experimental dataset we utilized includes different types of oil spill SAR images, and the experimental evaluations from different metrics for the implemented segmentation techniques are given in Sections V-C–V-F.

TABLE I
Satellite Sensors and Image Description

| Satellite Sensor | Spatial Resolution | Waveband | Image Level |
|---|---|---|---|
| Radarsat-SAR | 50m x 50m | C-band | 2 |
| ERS-1 SAR | 30m x 30m | C-band | 2 |
| ERS-2 SAR | 30m x 30m | C-band | 2 |
| Envisat ASAR | 150m x 150m | C-band | 2 |

TABLE II
Description of Oil Spill Images

| Capture Time | Satellite Sensor | Type of Oil Spills | Image Cover Ground |
|---|---|---|---|
| 08.05.2010 12:01:25 | Radarsat | - | 207.4 $km^2$ |
| 19.06.1995 02:30:55 | ERS-1 | - | 394 $km^2$ |
| 19.06.1995 02:31:09 | ERS-1 | - | - |
| 18.07.1998 02:03:24 | ERS-2 | Ship Oil Spill | - |
| 17.03.2000 01:54:37 | ERS-2 | Ship Oil Spill | - |
| 26.07.1997 02:26:34 | ERS-2 | - | 17.8x106$m^2$ |
| 01.09.2008 01:11:51 | Envisat | Ship Oil Spill | - |
| 15.08.2007 13:04:03 | Envisat | - | - |
| 28.07.2008 12:26:30 | Envisat | - | - |

### A. Experimental Preparations

*Image dataset:* To evaluate the segmentation performance of the proposed method, in the experimental work, we conduct experimental evaluations for the proposed SRCNet using image dataset obtained from the NOWPAP database,[1] and the dataset contains different types of oil spill SAR images. Specifically, the image data utilized in the experimental work includes SAR and ASAR images with VV, VH, and HH polarization. These images were captured by different sensors at different time, and the oil spills in the experimental images exhibit various areas with irregular shapes. In particular, the image dataset utilized includes C-band oil spill SAR images from ERS-1 and ERS-2 satellites, C-band oil spill SAR images from Radarsat satellite, and C-band oil spill ASAR images from ENVISAT satellite. We provide the specific descriptions for the experimental image dataset in Tables I and II, where the symbol "-" in Table II indicates unavailable information. In the experimental implementation, the dataset we utilized contains 1200 image samples, and these samples are split in training, validation, and testing sets. Specifically, we use 70% of the dataset for training, 10% for validation, and the remaining 20% of the dataset for testing.

[1] http://cearac.poi.dvo.ru/en/db/

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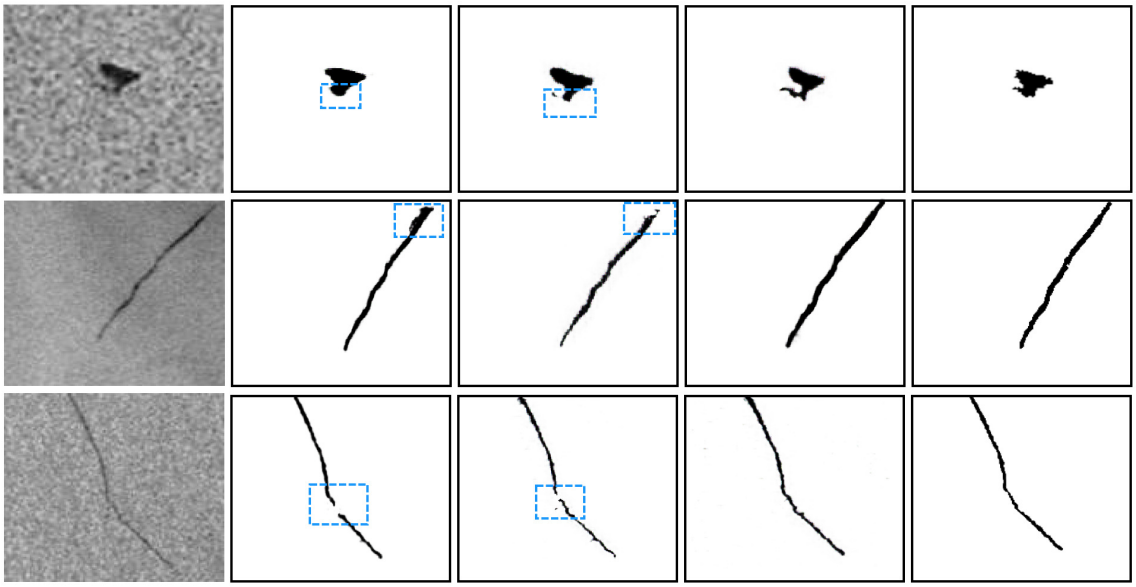


Fig. 3. Segmentation of oil spill SAR images with the exploitation of different segmentation techniques. Specifically, (a) Original oil spill images, and from top to bottom are the ERS-1 SAR, ERS-2 SAR, and ENVISAT ASAR oil spill images, respectively. (b)–(d) Corresponding segmentation results of LSE, RAC, and our proposed method, respectively. (e) Ground-truth segmentations. The blue dashed-line boxes are utilized to indicate the incorrect segmentations.

### *B. Experimental Setup*

To demonstrate the effectiveness of the proposed SRCNet for oil spill SAR image segmentation, we conduct experimental evaluations on different types of oil spill SAR images, and the specifications for image data information are given in Section V-A. The details of implementation training for oil spill SAR image segmentation are given as follows.

*1) Implementation Details:* In the experimental work, we conduct the training for the proposed segmentation network with the input SAR images scaled to the size of $128 \times 128$. In the training dataset, the segmentation masks are also provided, which indicate the real oil spill areas clearly, and thus using the segmentation masks as a criterion of training the segmentation network helps to empower its capability in terms of correctly segmenting oil spills in SAR images.

*2) Training Operation:* In the training of our proposed SRCNet for oil spill SAR image segmentation, we conduct the training work using the Tensorflow framework. Specifically, the training is conducted using Adam optimizer [53] with the learning rate set to $\alpha = 1e-4$, and the segmentation network is trained on a high performance computing sever with a NVIDIA Tesla P100 GPU with 16 GB memory for 180 epochs, and the batch size is set as 1.

*3) Experimental Evaluation Metrics:* In the experimental work, we conduct experimental evaluations from different metrics. According to [54] and [55], the segmentation accuracy is a critical evaluation criterion for image segmentation techniques. Thus, we evaluate the segmentation performance of the proposed segmentation network with respect to segmentation accuracy computed as follows:

$$\text{Accuracy} = \frac{\#\text{ the number of correctly segmented pixels}}{\#\text{ the number of all pixels}}. \tag{17}$$

TABLE III
SEGMENTATION ACCURACY OF LSE, RAC, AND OUR PROPOSED METHOD

| Image \ Method | LSE | RAC | Our Method |
|---|---|---|---|
| ERS-1 SAR | 0.8659 | 0.9031 | **0.9409** |
| ERS-2 SAR | 0.9319 | 0.9308 | **0.9869** |
| Envisat ASAR | 0.9209 | 0.9386 | **0.9907** |

Moreover, in the operation of image segmentation, the Jaccard index (JCI) is regarded as an useful metric in evaluating the segmentation performance [56], [57]. Thus, to make further evaluation for the proposed method in terms of operating oil spill segmentation in SAR images, we conduct the segmentation evaluation with respect to JCI, which is given as follows:

$$\text{JCI} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{18}$$

where TP, FP, and FN represent the correctly segmented oil spill pixels, the background pixels that are incorrectly segmented as oil spills, and the oil spill pixels that are incorrectly segmented as background, respectively. Thus, JCI is the intersection of the actual oil spill areas and the segmented areas, divided by the union of these two sections, and a higher value of JCI represents a more accurate oil spill segmentation.

### *C. Comparison With Manual Assistant Segmentation Methods*

In this work, we propose an effective oil spill SAR image segmentation network named SRCNet, which performs oil

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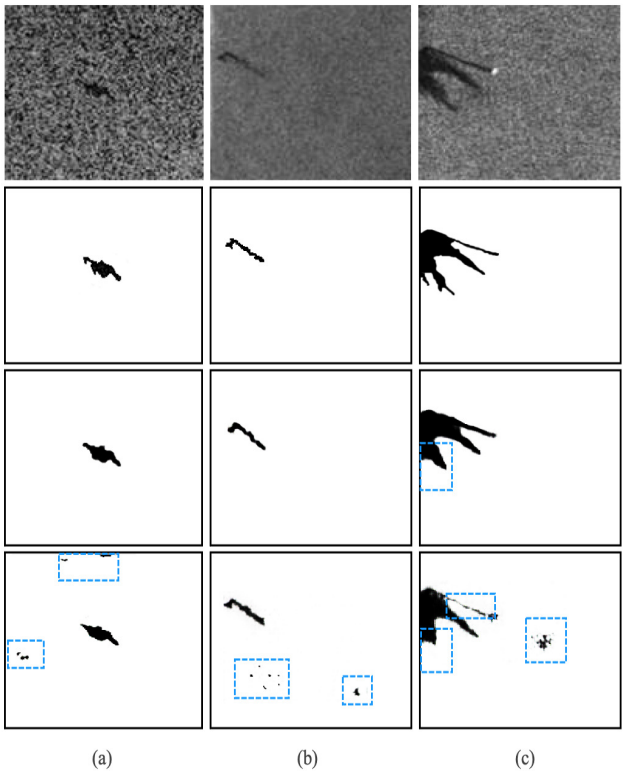


Fig. 4. Oil spill SAR image segmentation with the exploitation of our proposed segmentation network and the generative adversarial network. Specifically, in columns (a)–(c) from top to bottom are the original oil spill SAR image, the ground-truth segmentation, and the segmentation results of our proposed SRCNet and GAN, respectively. The blue dashed-line boxes are utilized to indicate the incorrect segmentations.

spill segmentation automatically. Thus, to evaluate the segmentation performance of our proposed SRCNet, in this part, we conduct the experimental evaluation by comparing its segmentation against several representative manual assistant segmentation methodologies. These comparison methodologies include level set evolution (LSE) [58] strategy and the region active contour (RAC) model [59]. These two manual assistant segmentation methods operate oil spill image segmentation requiring a starting region that is manually labeled to start the segmentation process, and thus the segmentation outcomes are correlated to the starting region to some extent. In this scenario, taking them as comparison provides a clear observation for the performance of our proposed segmentation network. Particularly, to conduct a more detailed evaluation for the segmentation performance of our proposed SRCNet, we here utilize different types of oil spill SAR images as the experimental test dataset, and the segmentation results are shown in Fig. 3. In this figure, (a) from top to bottom are the original ERS-1 SAR, ERS-2 SAR, and the ENVISAT ASAR oil spill images, respectively, (b)–(d) are the corresponding segmentation results of the LSE, RAC, and our proposed segmentation method, respectively, and (e) shows the ground-truth segmentations. Examining the segmentation results, it is clear that our proposed SRCNet achieves a more accurate oil spill SAR image segmentation.

Additionally, in the implementation of image segmentation, the metric of segmentation accuracy represents pixel-wise

TABLE IV
SEGMENTATION ACCURACY OF GAN AND OUR PROPOSED METHOD

| Image \ Method | GAN | Our Method |
|---|---|---|
| ERS-1 | 0.8903 | **0.9725** |
| ERS-2 | 0.9271 | **0.9910** |
| Radarsat | 0.7993 | **0.9109** |

TABLE V
SEGMENTATION ACCURACY OF SFE, MGS, AND OUR PROPOSED METHOD

| Image \ Method | SFE | MGS | Our Method |
|---|---|---|---|
| Radarsat SAR | 0.9254 | 0.9269 | **0.9904** |
| ERS-1 SAR | 0.9209 | 0.9231 | **0.9881** |
| ERS-2 SAR | 0.9683 | 0.9709 | **0.9897** |

evaluation in terms of measuring the percentage of correctly segmented pixels [56]. Thus, to quantitatively evaluate the segmentation performance of our proposed method, we further compute its segmentation with respect to segmentation accuracy, and the computation results are shown in Table III. In this table, we can see that our proposed method achieves higher segmentation accuracy compared against the LSE and the RAC segmentation methods. This benefits from the fact that our proposed SRCNet operates automatic oil spill SAR image segmentation without requiring manually setting a starting region to implement the segmentation process. Particularly, turbulence and wind blowing on the ocean surface normally result in oil spills in SAR images exhibiting various areas with irregular shapes, and thus it is not easy to capture a proper region for starting oil spill segmentation. Besides, the manually setting captures the segmentation starting region coarsely, which is not reliable for accurate oil spill segmentation. Therefore, compared with the manual assistant segmentation methods, our proposed SRCNet achieves more effective oil spill image segmentation.

### *D. Comparison With the GAN*

As described in the application of deep neural networks for image processing, a striking point is the involvement of discriminative modules which learn to determine whether an example is from the formulated model distribution or the real image data so as for increasing the overall capability of the networks [60]. The generative adversarial net (GAN) [61] is a representative deep neural network, in which a discriminative module is structured to estimate the probability that a sample comes from training image data rather than the generated one. Therefore, to further evaluate the segmentation performance

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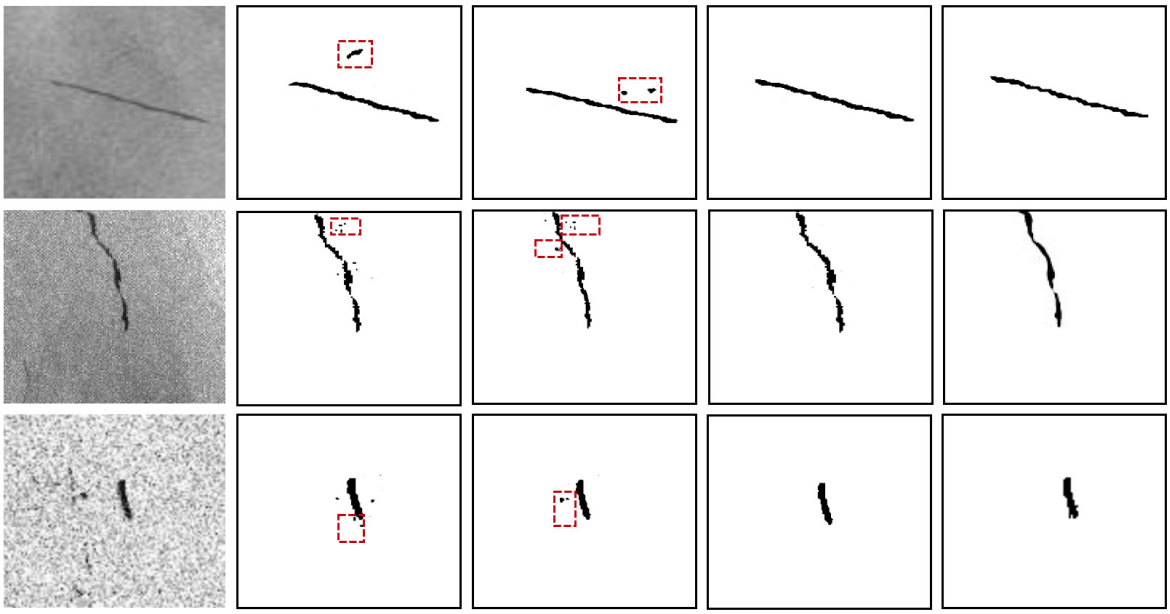


Fig. 5. Segmentation for oil spill SAR images with the exploitation of the proposed segmentation method and the other state-of-the-art segmentation networks. Specifically, (a) Original oil spill SAR images, and from top to bottom are the Radarsat, ERS-1, and ERS-2 oil spill SAR images, respectively. (b)–(d) Corresponding segmentation results of SFE, MGS, and our proposed segmentation method, respectively. (e) Ground-truth segmentations. The red dashed-line boxes are utilized to indicate the incorrect segmentations.

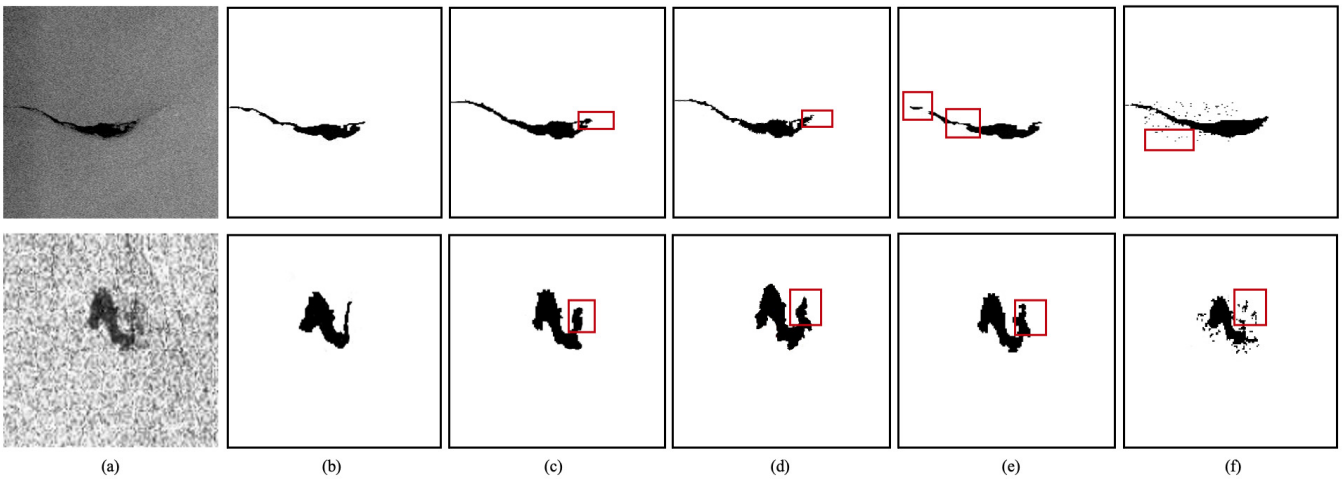


Fig. 6. Comprehensive evaluation for the proposed segmentation method for oil spill SAR image segmentation. Specifically, (a) Original oil spill SAR images. (b) Segmentation masks. (c)–(f) Corresponding segmentation results of the proposed segmentation method, DGNet, ISPDM, and VLSA, respectively. The red boxes are utilized to indicate the incorrect segmentations.

of our proposed method in terms of segmentation oil spills from SAR images, in this part, we conduct the experimental work by comparing its segmentation performance with GAN. Specifically, to examine the segmentation performance of our proposed SRCNet and GAN for segmentation different types of oil spill images, we here exploit ERS-1, ERS-2, and Radarsat oil spill SAR images as our experimental dataset, and the segmentation results are shown in Fig. 4. In this figure, the first row from left to right are the ERS-1, ERS-2, and Radarsat oil spill SAR images, respectively, and in each column from top to bottom are the original oil spill SAR image, the ground-truth segmentation, and the segmentations of our proposed segmentation network and GAN, respectively. Comparing the segmentation results of our proposed method and GAN with the ground-truth segmentation, it is obvious that our proposed method achieves more accurate oil spill image segmentation.

Additionally, to further evaluate the segmentation performance of our proposed segmentation network, we validate its segmentation with respect to segmentation accuracy shown in Table IV. Examining the segmentation results in this table, it is clear that our proposed SRCNet achieves higher segmentation accuracy compared against GAN.

### *E. Comparison With State-of-the-Art Segmentation Technique*

To evaluate the segmentation performance of the proposed segmentation network one step further, in this part, we conduct the experimental evaluation by comparing its segmentation performance with other alternative state-of-the-art

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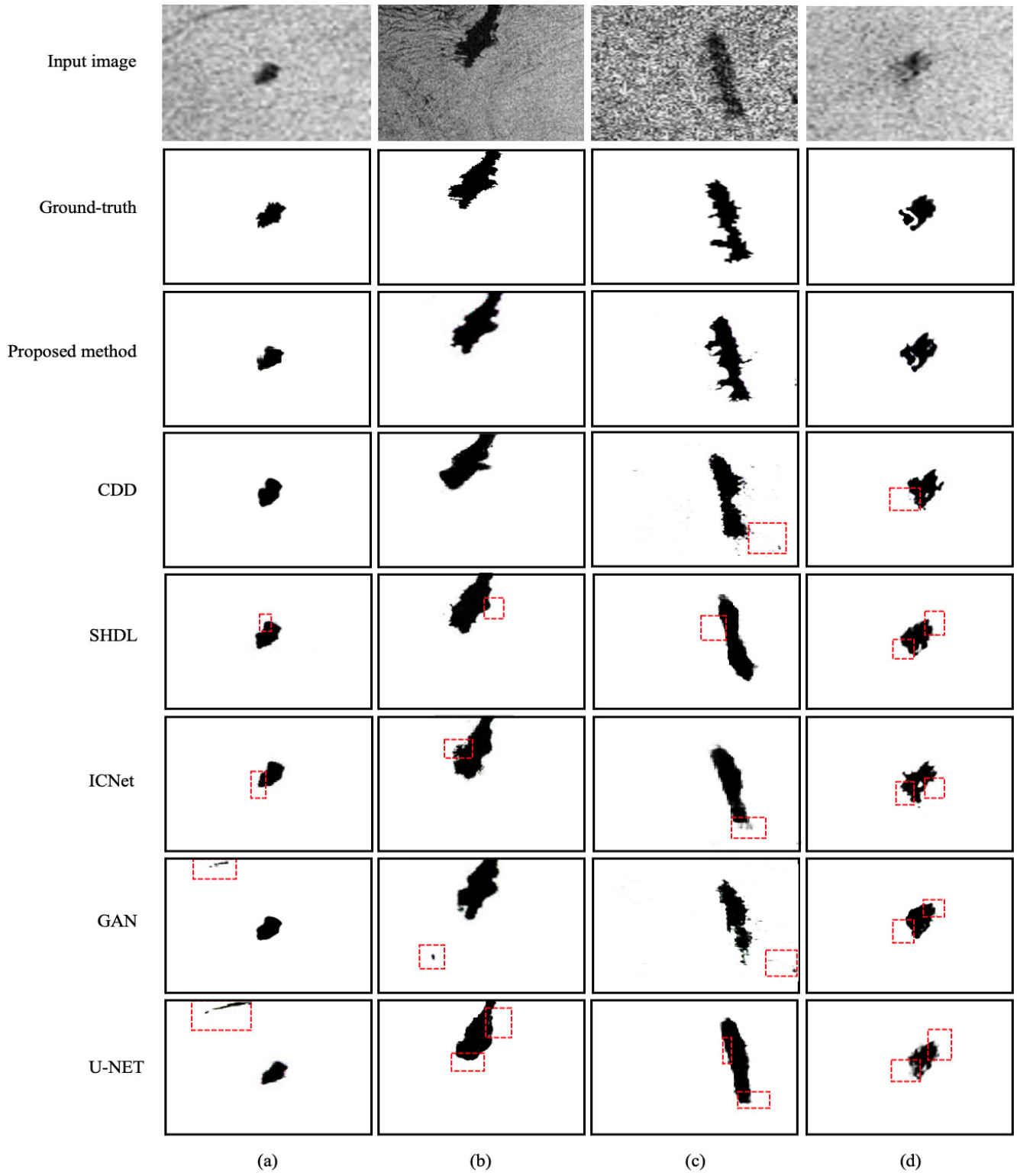


Fig. 7. Segmentation of ERS-1 oil spill SAR images with the exploitation of different segmentation methodologies. (a)–(d) (Top to bottom) Oil spill SAR images and their corresponding segmentation results using the implemented segmentation techniques. Particularly, in this figure, from top to bottom in each column are the original oil spill SAR image, the ground-truth segmentation, the segmentation results of our proposed SRCNet, CDD, SHDL, ICNet, GAN, and tU-Net segmentation methodologies, respectively. The red dashed-line boxes are utilized to indicate the incorrect segmentations.

segmentation techniques. Specifically, the employed comparison segmentation techniques include SFE [38] and MGS [39] segmentation networks. The segmentation results are shown in Fig. 5. In this figure, from left to right are the original oil spill SAR images, the segmentation results of SFE, MGS, and the proposed segmentation method separately, and the ground-truth segmentations. Examining this figure, it is clear that the proposed segmentation network achieves more accurate

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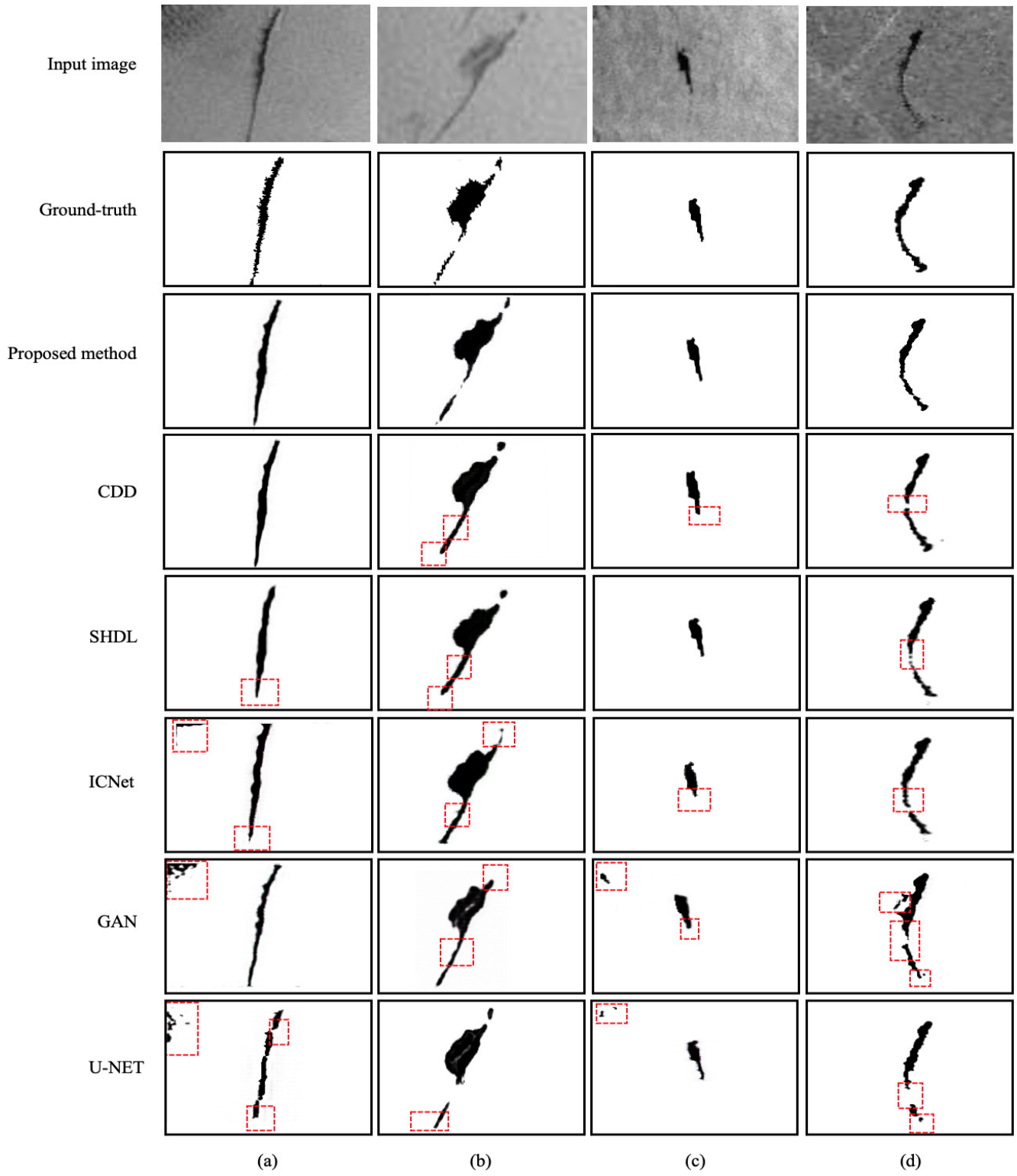


Fig. 8. Segmentation of ERS-2 oil spill SAR images with the exploitation of different segmentation methodologies. (a)–(d) (Top to bottom) Oil spill SAR images and their corresponding segmentation results using the implemented segmentation techniques. Particularly, in this figure, from top to bottom in each column are the original oil spill SAR image, the ground-truth segmentation, and the segmentation results of our proposed SRCNet, CDD, SHDL, ICNet, GAN, and U-Net segmentation methodologies, respectively. The red dashed-line boxes are utilized to indicate the incorrect segmentations.

oil spill segmentation compared against the other employed state-of-the-art segmentation techniques. Besides, to further evaluate the performance of the proposed method, we quantitatively examine its segmentation performance with respect to the segmentation accuracy, and the results are given in Table V. From this table, it is clear that the proposed method achieves higher segmentation accuracy compared against the SFE and the MGS segmentation methods. The evaluations

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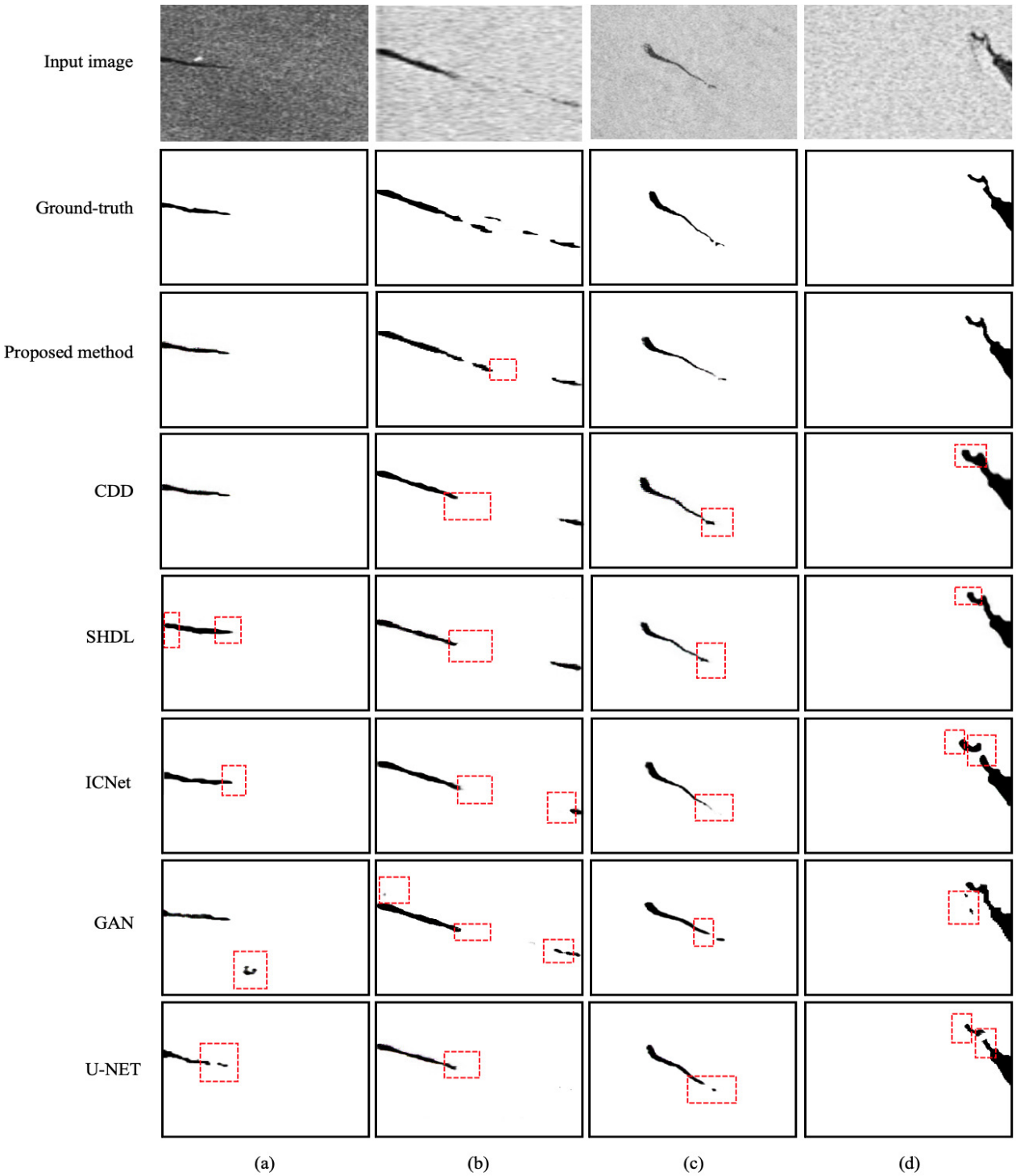


Fig. 9. Segmentation of Radarsat oil spill SAR images with the exploitation of different segmentation methodologies. (a)–(d) (Top to bottom) Oil spill SAR images and their corresponding segmentation results using the implemented segmentation techniques. Particularly, in this figure, from top to bottom in each column are the original oil spill SAR image, the ground-truth segmentation, the segmentation results of our proposed SRCNet, CDD, SHDL, ICNet, GAN, and U-Net segmentation methodologies, respectively. The red dashed-line boxes are utilized to indicate the incorrect segmentations.

from both the qualitative and the quantitative metrics validate the effectiveness of the proposed method for oil spill SAR image segmentation.

### *F. Comprehensive Evaluation for Oil Spill Image Segmentation*

We evaluated the segmentation performance of the proposed SRCNet for oil spill image segmentation by comparing its segmentation with the manual assistant segmentation methods, the GAN, and the state-of-the-art segmentation techniques in Sections V-C–V-E, respectively, and the experimental results demonstrated the effectiveness of the proposed method. To evaluate the segmentation performance of the proposed method one step further, in this part, we conduct the experimental evaluation for its performance comprehensively. Specifically, we commence the experimental evaluation by

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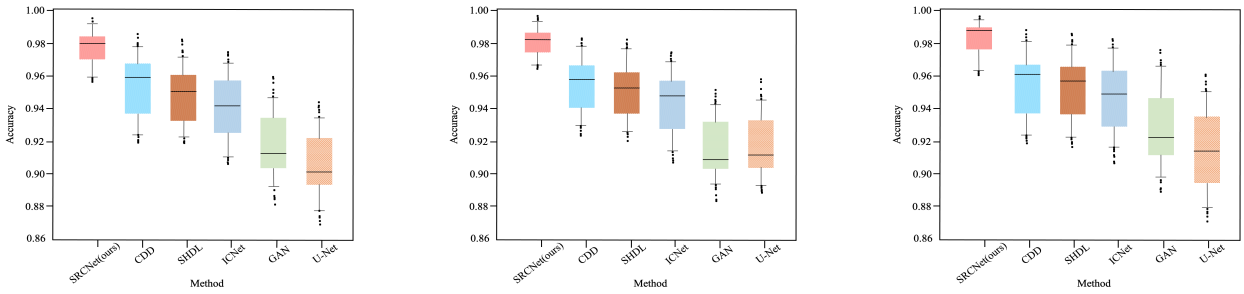
TABLE VI
SEGMENTATION ACCURACY OF ERS-1 OIL SPILL SAR IMAGES

| Image \ Method | U-Net | GAN | ICNet | SHDL | CDD | SRCNet (Ours) |
|---|---|---|---|---|---|---|
| (a) | 0.8958 | 0.9307 | 0.9682 | 0.9703 | 0.9728 | **0.9794** |
| (b) | 0.9026 | 0.9458 | 0.9473 | 0.9582 | 0.9831 | **0.9869** |
| (c) | 0.9021 | 0.8997 | 0.9228 | 0.9231 | 0.9308 | **0.9847** |
| (d) | 0.8923 | 0.9037 | 0.9249 | 0.9204 | 0.9385 | **0.9731** |




(a) (b) (c)

Fig. 10. Segmentation accuracy distributions of the implemented segmentation methods for different types of oil spill image segmentation. Specifically, (a)–(c) distributions of the segmentation accuracy that correspond to each segmentation method for ERS-1, ERS-2, and Radarsat oil spill image segmentation, respectively. Particularly, in each sub-figure, the rectangles corresponding to the segmentation strategies depict the accuracy values distributed between the minimum and the maximum values (these two values are shown with bars on the bottom-up sides of the rectangles), and the black dots are the segmentation outliers that fall out the interval of the two bars.

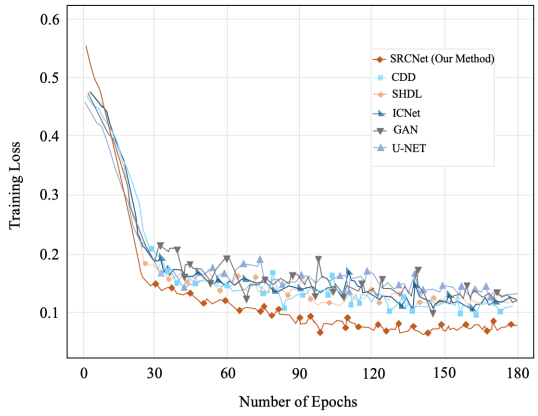



Fig. 11. Learning curve of our proposed oil spill image segmentation network and the other implemented segmentation techniques.

comparing the segmentation performance of the proposed method with the DGNet [51] segmentation network, the ISPDM [62] segmentation strategy, and the variational level set approach (VLSA) [63]. In particular, the DGNet and the ISPDM are state-of-the-art oil spill image segmentation techniques, and the VLSA is a classic image segmentation approach. In the segmentation implementation, the DGNet operates oil spill segmentation in SAR images automatically as it was developed based on deep learning, and the ISPDM performs oil spill SAR image segmentation via probability distribution modeling. Therefore, using the DGNet, ISPDM, and VLSA for comparison enables a comprehensive evaluation for the performance of the proposed segmentation method, and the segmentation results are illustrated in Fig. 6. Examining the results in this figure, it is clear that the proposed segmentation method achieves a more accurate oil spill SAR image segmentation compared against the other employed segmentation methods. This demonstrates the superiority of the proposed method for oil spill SAR image segmentation.

Besides, to further evaluate the segmentation performance of the proposed segmentation method, we conduct the experimental evaluation for the proposed SRCNet using different evaluation metrics. This enables a more comprehensive examination for the performance of the proposed SRCNet for segmentation oil spill SAR images. Specifically, to operate a detailed evaluation for the proposed method, we conduct the experimental evaluation by comparing its segmentation against several representative image segmentation techniques using ERS-1, ERS-2, and Radarsat oil spill SAR images as the experimental dataset, and the segmentation results are shown in Figs. 7–9. In detail, the comparison segmentation methods utilized in this experimental part include capacitory discrimination distance (CDD) learning technique [64], squared Hellinger divergence learning (SHDL) [64] segmentation model, image cascade network (ICNet) for image segmentation [65], GAN, and U-Net segmentation network [66]. Specifically, Figs. 7–9 show the segmentation for ERS-1, ERS-2, and Radarsat oil spill SAR images, respectively, using

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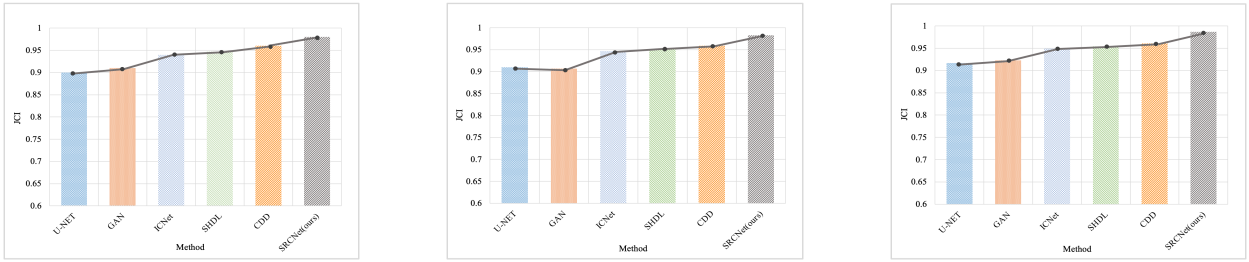


(a) (b) (c)

Fig. 12. Segmentation performance of the employed segmentation networks for oil spill SAR image segmentation with respect to JCI. (a)–(c) JCI of the employed segmentation methods for ERS-1, ERS-2, and Radarsat oil spill SAR image segmentation, respectively, and the columns in each sub-figure indicate the levels of the computed JCI for the employed segmentation networks. Particularly, the dark points on the top of the columns are utilized to mark the levels of the computed JCI of the employed segmentation methods, and the lines that connect the dark points show the trend of the segmentation performance of the employed alternative segmentation methods.

our proposed SRCNet and the comparison methodologies. Particularly, in each figure, from top to bottom are the original oil spill SAR images, the ground-truth segmentation, and the segmentation results of our proposed segmentation network, CDD, SHDL, ICNet, GAN, and U-Net, respectively. Examining the segmentation results, it is obvious that our proposed SRCNet performs a more accurate oil spill SAR image segmentation compared against the other employed segmentation methods.

Besides the qualitative segmentation evaluation, to further validate the segmentation performance of the proposed SRCNet and the other employed segmentation methodologies for oil spill image segmentation, we here conduct the quantitative experimental evaluation in terms of the segmentation accuracy shown in Tables VI–VIII. Specifically, Tables VI–VIII show the segmentation accuracy of the implemented segmentation techniques (including the proposed SRCNet) for ERS-1, ERS-2, and Radarsat oil spill SAR image segmentation, respectively. Examining the experimental results in these tables, it is obvious that our proposed segmentation method achieves higher segmentation accuracy than the other employed segmentation techniques.

The segmentation results with respect to qualitative evaluation shown in Figs. 7–9 and the quantitative experimental evaluation in terms of segmentation accuracy shown in Tables VI–VIII demonstrate the effectiveness of our proposed SRCNet for different types of oil spill SAR image segmentation. Together with these segmentation evaluations, to further validate the segmentation performance of the proposed segmentation network, we evaluate its overall segmentation performance statistically. The statistical evaluation performed here is capable of simultaneously demonstrating both the overall segmentation accuracy and the segmentation stability of the employed segmentation methodologies quantitatively, where box plots are utilized, and the experimental results are illustrated in Fig. 10. Specifically, Fig. 10(a)–(c) demonstrates the distribution ranges of the statistical computation for the segmentation accuracy with respect to the implemented segmentation strategies for ERS-1, ERS-2, and Radarsat oil spill SAR image segmentation, respectively. In particular, the colored boxes in each sub-figure are utilized to depict the clustered accuracy values, where the bars on the bottom-up sides of the boxes indicate the levels of the minimum and the maximum values, and the black dots are the segmentation outliers that fall out the interval of the two bars. Examining the statistical computation results, it is obvious that our proposed segmentation network achieves higher segmentation accuracy for oil spill SAR image segmentation with a smaller number of outliers. This demonstrates that our proposed SRCNet achieves more accurate and stable oil spill SAR image segmentation compared against the other implemented segmentation methodologies.

Our proposed SRCNet operates oil spill image segmentation by leveraging seminal image representation to empower its learning capability for accurate oil spill segmentation. The seminal image representation collaborated in the proposed SRCNet provides pixel-wise physical information for characterization oil spill SAR image data. Thus, in the training process, the collaborated seminal image representation empowers effective learning for accurate oil spill segmentation. To demonstrate the learning efficiency of the proposed SRCNet, we illustrate the learning curve of the employed segmentation networks shown in Fig. 11. In this figure, it is clear that our proposed SRCNet converges to a lower training loss with less epochs compared against the other implemented segmentation networks, and with the training epoch increasing, our proposed segmentation network fine-tunes the segmentation map and thus achieves accurate oil spill SAR image segmentation.

Additionally, in the implementation of oil spill SAR image segmentation, an accurate segmentation is that the segmented oil spill areas tightly approach the ground-truth segmentations. The closeness between the segmented oil spill areas and the ground-truth segmentations is mathematically described as the intersection. Moreover, in the operation of deep neural networks for image segmentation, the JCI is a key metric that indicates the segmentation performance of the employed segmentation networks [56], [67]. In this regard, to further evaluate the segmentation performance of the implemented segmentation networks (including our proposed SRCNet), we demonstrate the computed JCI of each segmentation network for oil spill SAR image segmentation shown in Fig. 12.

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TABLE VII
SEGMENTATION ACCURACY OF ERS-2 OIL SPILL SAR IMAGES

| Image \ Method | U-Net | GAN | ICNet | SHDL | CDD | SRCNet (Ours) |
|---|---|---|---|---|---|---|
| (a) | 0.8959 | 0.8981 | 0.9254 | 0.9683 | 0.9721 | **0.9883** |
| (b) | 0.9268 | 0.9145 | 0.9196 | 0.9239 | 0.9247 | **0.9893** |
| (c) | 0.9521 | 0.9430 | 0.9731 | 0.9828 | 0.9821 | **0.9908** |
| (d) | 0.9233 | 0.9286 | 0.9679 | 0.9683 | 0.9668 | **0.9901** |

TABLE VIII
SEGMENTATION ACCURACY OF RADARSAT OIL SPILL SAR IMAGES

| Image \ Method | U-Net | GAN | ICNet | SHDL | CDD | SRCNet (Ours) |
|---|---|---|---|---|---|---|
| (a) | 0.9685 | 0.9507 | 0.9723 | 0.9782 | 0.9886 | **0.9923** |
| (b) | 0.8731 | 0.9054 | 0.9003 | 0.9238 | 0.9221 | **0.9647** |
| (c) | 0.9623 | 0.9689 | 0.9726 | 0.9808 | 0.9821 | **0.9897** |
| (d) | 0.9543 | 0.9657 | 0.9639 | 0.9743 | 0.9725 | **0.9931** |

Specifically, Fig. 12(a)–(c) illustrates the computation results of JCI for ERS-1, ERS-2, and Radarsat oil spill SAR image segmentation with the utilization of the employed alternative segmentation networks, respectively, and the columns in each sub-figure indicate the levels of the computed JCI. From the description and the computation representation for JCI that a higher level value represents a more accurate oil spill image segmentation. For clarity, we mark the levels of the computed JCI with dark points and connect these points with lines to show the trend of the segmentation performance for the employed alternative segmentation methods. Examining the columns in each sub-figure, it is clear that our proposed SRCNet achieves a higher level of JCI compared against the other segmentation techniques. This demonstrates that the segmented oil spill areas from our proposed SRCNet tightly approach the ground-truth segmentations, achieving accurate oil spill segmentation.

## VI. CONCLUSION

We have presented the procedures of constructing our proposed SRCNet for oil spill image segmentation. The proposed SRCNet performs accurate oil spill image segmentation automatically. Specifically, in our proposed SRCNet, two deep neural nets are structured that work in a competitive manner, where one is the mapped generative net, which strives to produce accurate oil spill segmentation maps by drawing samples from the seminal image distribution representation, while the other is the discriminative net, which aims to distinguish the generated segmentation from the real oil spill segmentation. The competition enhances the segmentation performance of the proposed SRCNet for oil spill image segmentation. In the training process, the segmentation network reaches convergence when the generated segmentation is as accurate as the real oil spill segmentation, in which case the discriminative net is not able to distinguish between these two segmentations. In this regard, to render efficient and effective oil spill segmentation, we construct our proposed SRCNet with the collaboration of the seminal image representation. Specifically, the seminal image representation refers to the physical distribution representation of the pixels in SAR image, which models with the internal characteristics of oil spill SAR image data. Thus, in the training for oil spill image segmentation, the seminal image representation empowers the learning capability of the proposed segmentation network for producing accurate oil spill segmentation maps in high efficiency. This establishes an efficient and effective way for oil spill image segmentation. Besides, to further improve the segmentation performance of the proposed SRCNet, a regularization term that penalizes the segmentation loss is formulated, which encourages the produced segmentation to be close to the ground-truth segmentation, promoting the segmentation capability of the proposed SRCNet for a more accurate oil spill segmentation. Experimental evaluations from different metrics validate the effectiveness of the proposed SRCNet for accurate oil spill segmentation over different sources of oil spill images.

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